

Practice 5: M-QAM modulation

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https://github.com/santiagoUIS/GNURADIO_COMUIS_2025_2_B1_G7/tree/Practica_5

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Abstract—The analysis of modulated digital signals is fundamental to understanding the behavior of modern communication systems, especially in noisy environments where fidelity in information transmission is critical. In this context, M-QAM and M-PSK modulations allow the relationship between spectral efficiency and robustness against noise to be adapted, making them essential tools in the design of digital links.

This laboratory practice focused on the study of digitally modulated signals in their baseband and bandpass forms, using Software-Defined Radio (SDR). BPSK, QPSK, 8-PSK, and 16-QAM modulation schemes were implemented, analyzing their behavior in the presence of additive white Gaussian noise and observing the effects on the resulting constellations.

The power spectral density (PSD) of each signal was examined, comparing its theoretical and simulated characteristics, and the influence of conversion between baseband and passband on the spectral shape was studied. The results obtained allowed us to correlate performance degradation with increasing modulation order and noise level, reinforcing fundamental concepts about digital signal transmission and processing.

keywords: PSK, QPSK, 8-PSK, 16-QAM, Gaussian noise, Bandpass, Baseband, Spectral efficiency, Digital transmission.

I INTRODUCTION

M-QAM modulation is a digital technique that combines variations in the amplitude and phase of a carrier to transmit information. In M-QAM modulation, each symbol represents $\log_2(M)$ bits, where M is the total number of points in the constellation. For example, 4-QAM, also called QPSK, transmits 2 bits per symbol, 16-QAM transmits 4 bits, and 64-QAM transmits 6 bits per symbol, thus increasing the spectral efficiency of the system.

This technique is widely used in modern communication systems such as Wi-Fi, LTE/5G, cable modems (DOCSIS), and digital television, where a balance between transmission rate and robustness against noise

is sought. However, as the modulation order (M) increases, the constellation points move closer together, making the system more sensitive to noise and phase or amplitude errors, thus requiring a higher signal-to-noise ratio (SNR) to maintain an acceptable error rate.

M-QAM modulation is an efficient alternative to purely phase-based schemes such as M-PSK, as it allows more bits per symbol to be transmitted without increasing bandwidth, albeit at the cost of greater complexity and lower noise tolerance. Its use is essential in broadband communications and multichannel systems, where spectrum optimization is a priority.

In this practice, the behavior of modulated digital signals using **M-QAM** and **M-PSK** schemes was studied, analyzing their representation in **baseband** and their corresponding **bandpass** form. BPSK, QPSK, 8-PSK, and 16-QAM modulations were implemented in order to observe their differences in spectral efficiency and constellation structure. The influence of white Gaussian additive noise on each modulation scheme was evaluated, observing how the increase in noise power affects the dispersion of the points in the constellation and the probability of error. Finally, the PSD of the baseband and passband signals were analyzed, comparing their theoretical and experimental behavior.

The study provided insight into the relationship between the complexity of the modulation scheme, robustness against noise, and spectrum efficiency, which are fundamental aspects in the design of modern digital communication systems.

II GENERAL OBJETIVE

Learn how to program different types of digital modulation schemes based on signal constellations, such as PSK and QAM. This includes understanding the mathematical representation of constellation points, generating baseband and passband versions of modulated signals, and analyzing their spectral and temporal behavior.

III THEORETICAL FRAMEWORK

III-A M-QAM modulation

M-QAM combines amplitude modulation (AM) and phase modulation (PM) in the same scheme. The main idea is to represent each transmitted symbol as a point in the complex plane I - Q .

Each point in the **constellation** corresponds to a different symbol, and its position (in amplitude and phase) encodes a set of bits as shown in Fig. 1. In this way, the binary information is mapped onto coordinates in the complex plane, allowing multiple bits to be transmitted per symbol by jointly varying the amplitude and phase of the carrier.

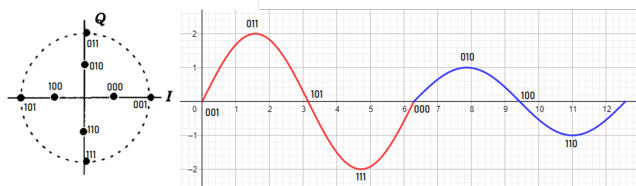


Figure 1. Complex plane representation and M-QAM modulation.

A M-QAM signal can be expressed as:

$$s(t) = I_k \cos(2\pi f_c t) - Q_k \sin(2\pi f_c t) \quad (1)$$

where:

- f_c is the carrier frequency.
- I_k and Q_k are the in-phase and quadrature amplitude components of the signal, respectively (discrete values depending on the transmitted symbol).
- k denotes the symbol index.

To demodulate, the received signal $s(t)$ in Eq. 1 is multiplied by the carrier components $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$, and then passed through low-pass filters to recover the in-phase and quadrature components I_k and Q_k , respectively[1].

This operation effectively shifts the signal from the passband to the baseband. When $s(t)$ is multiplied by $\cos(2\pi f_c t)$, the in-phase component I_k produces a constant (DC) term, while the quadrature component Q_k generates high-frequency terms around $2f_c$. A similar effect occurs when multiplying by $\sin(2\pi f_c t)$ for the Q_k component[1].

After applying low-pass filters (LPFs), the high-frequency components are removed, leaving only the baseband signals proportional to I_k and Q_k . These recovered components represent the original symbol coordinates in the constellation diagram, from which

the receiver determines the closest constellation point and reconstructs the transmitted bit sequence[1].

With these recovered values, the receiver determines the closest point in the constellation diagram and thereby reconstructs the transmitted bits.

III-B BASEBAND AND PASSBAND SIGNALS

A baseband signal represents the low-frequency version of a communication signal, containing all the information to be transmitted but centered around zero frequency. These signals are typically complex-valued, with in-phase and quadrature components, and are convenient for digital processing and simulation[2].

In contrast, a passband signal is the real-valued version suitable for physical transmission over the communication channel. It is obtained by modulating the baseband signal onto a carrier frequency f_c , effectively translating its spectrum from baseband to a higher frequency band centered around f_c [2].

Mathematically, the relationship between a complex baseband signal $s_{bb}(t)$ and its corresponding real passband signal $s_{pb}(t)$ is given by:

$$s_{pb}(t) = \Re \{ s_{bb}(t) e^{j2\pi f_c t} \} \quad (2)$$

where $s_{bb}(t) = I(t) + jQ(t)$ is the complex envelope of the signal and $\Re\{\cdot\}$ denotes the real part. Expanding Eq. 2, the passband signal can be written as:

$$s_{pb}(t) = \frac{I(t) \cos(2\pi f_c t)}{2} - \frac{Q(t) \sin(2\pi f_c t)}{2} \quad (3)$$

This expression shows that the in-phase and quadrature components modulate two orthogonal carriers, resulting in a real-valued passband signal that carries both amplitude and phase information as shown in Fig. 2.

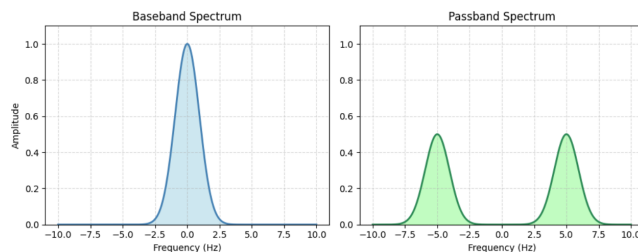


Figure 2. Example of a signal in its baseband and passband forms.

IV METHODOLOGY

This section presents in an orderly and detailed manner the experiments carried out during the laboratory, including simulations and other considerations addressed within the practice.

IV-A BANDBASE MODULATION IN M-QAM.

Initially, a general study of the flowchart in Fig. 3 is carried out, which is used to perform different M-QMA modulations.

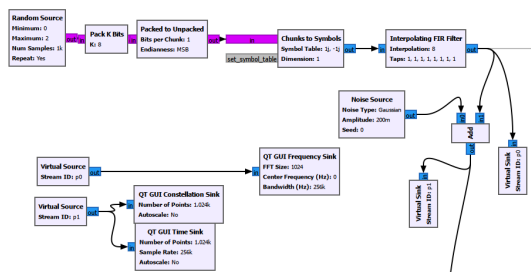


Figure 3. Initial schematic of the flowchart used to modulate base-band signals in M-QAM.

The transmitted signal is generated from a binary sequence produced by the **Random Source** block, which outputs a stream of random bits representing the digital information to be transmitted. This bit stream is then processed by the **Pack K Bits** block, which groups the bits into packets of size K , facilitating their subsequent conversion into symbols[3].

Next, the **Unpacked to Packed** block reorders or converts the bits into a suitable format for manipulation by the **Chunks to Symbols** block, where symbol mapping is performed according to the chosen modulation constellation[3].

Subsequently, the **Interpolating FIR Filter** block performs pulse shaping using a Finite Impulse Response filter, with an interpolation factor sps that determines the number of samples per symbol. This process allows control of the signal bandwidth and reduces Inter-Symbol Interference, ensuring a smoother pulse shape[3].

The necessary variables used in the flowchart are compiled, for which the following information is obtained:

- R_s : Symbol rate.
- R_b : Bit rate.
- M : Constellation table size.
- bps : Number of bits per symbol.
- sps : Symbols per second.

Subsequently, the correlation between them is determined, for which purpose the TABLE 1 is created.

IV-B PASSBAND MODULATION IN M-QAM.

For modeling the bandbase signal in bandpass filters, a new module is included that is connected directly to

predefined parameters	dependent parameters
R_s	$R_b = R_s \cdot bps$
SPS	$bps = \log_2(M)$
M	$f_s = R_s \cdot SPS$

Table 1

RELATIONSHIP BETWEEN SELECTABLE PARAMETERS AND THEIR EFFECT ON DEPENDENT PARAMETERS

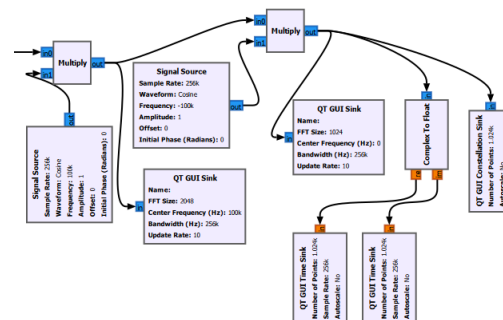


Figure 4. Schematic designed to convert a baseband signal to its bandpass form with added noise.

the output of the bandbase module with added noise. The designed module can be seen in Fig. 4.

For its respective implementation, Eq. 3 is used as a starting point. A sinusoidal signal source with cutoff frequency f_c is implemented to shift the center frequency of the PSD. Subsequently, a **QT GUI Sink** block is added to observe the modulated signal in both spectrum and frequency. To verify the proper functioning of the flowchart, baseband demodulation is performed again from a sinusoidal source but with a cutoff frequency $-f_c$. Additionally, the baseband signal is decomposed into its imaginary and real components to easily compare the original baseband signal.

V EXPERIMENTS AND RESULTS

Once the schematics for the bandabase and bandpass filters are available, the respective simulations are performed to observe the behavior of the signal over time, in frequency, and its respective constellation representation.

First, the flowchart proposed in Fig. 3 is used, temporarily disabling the bandpass modulator in Fig. 4. The result obtained can be seen in Fig. 5, highlighting the added noise with an amplitude of 0.1 [V] and its evident effect on the signal in the time domain and its respective quadrature. It should be noted that the robustness of BPSK and QPSK modulation against Gaussian noise is evident. On the other hand, the high susceptibility of much more spectrally efficient modulations such as 8-PSK or 16-QAM can be seen, where symbol interception

and, therefore, error are evident. In addition, particularities such as bandwidth conservation are appreciated for each modulation case, making it easy to identify a point of comparison for spectral efficiency. The bandwidth obtained is $\approx 60\text{kHz}$, defined by the relation $\frac{B}{2} = R_s$.

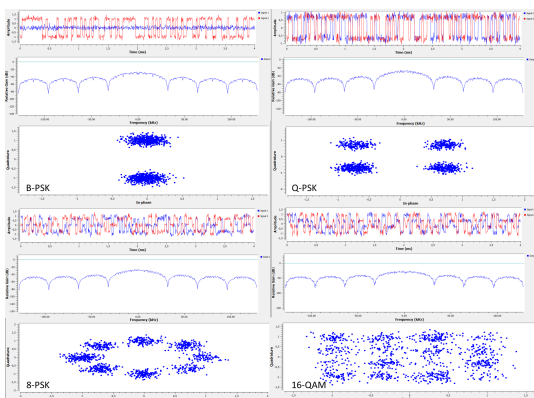


Figure 5. Output of the flowchart for bandbase signal processing. The image is divided into four quadrants showing the behavior in time, frequency, and constellation respectively of each modulation.

A $f_c = 100\text{kHz}$ is defined as the center frequency for the passband signal. The output of the flowchart can be seen in Fig. 6, which shows the response over time for the different modulations. First, the demodulated bandpass signal is observed, decomposed into its real and imaginary components over the original signal, in order to verify that the modulation is performed correctly. Next, the time signal modulated at the previously defined frequency f_c is shown.

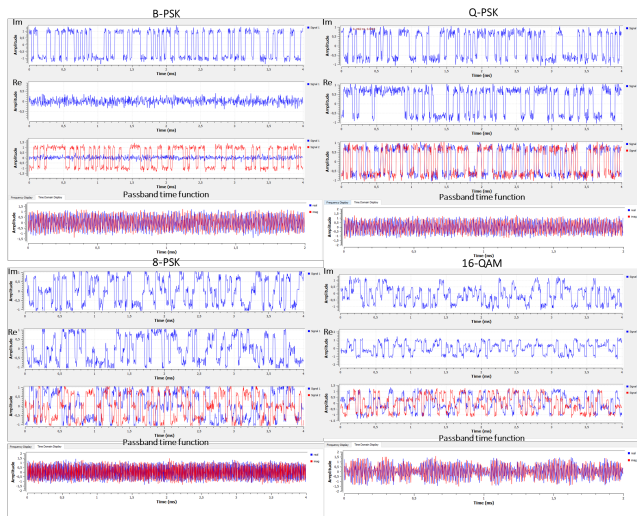


Figure 6. Passband signal processing divided into four quadrants showing the behavior in time of each modulation.

Note that for each modulation case, the baseband signal over time shows changing and abrupt behavior, since it now consists of frequencies between 70kHz and 130kHz. In addition, magnitude levels can be easily seen in 8PSK, and 16-QAM modulations.

In Fig. 7, it is possible to observe the PSD of the original band-modulated signal, followed by the PDS of the signal in its passband form and, finally, its respective representation in constellation.

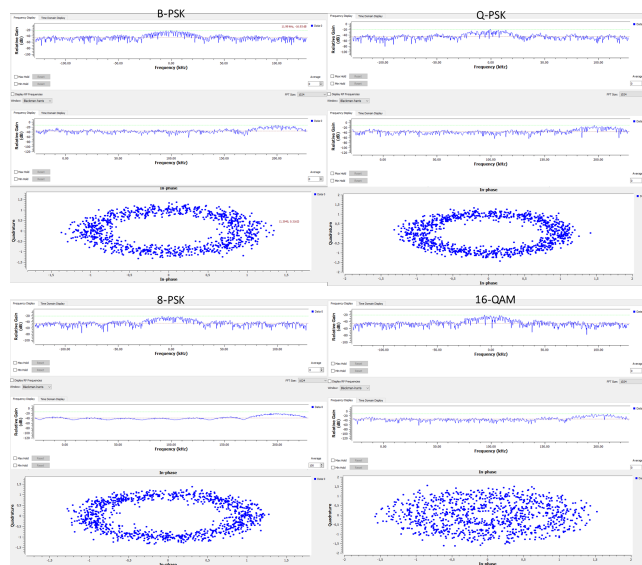


Figure 7. Passband signal processing divided into four quadrants showing the behavior in frequency and constellation of each modulation.

However, the information provided by the constellations is almost nil, since it is not possible to identify exact points of representation for each symbol. However, once the passband signals are brought to baseband, the constellation is recovered and is exactly the same as the constellation obtained during baseband signal processing.

VI CONCLUSIONS

- The analysis of M-QAM and related digital modulation schemes demonstrates that increasing the modulation order M directly enhances spectral efficiency, as more bits are transmitted per symbol. However, this improvement comes at the cost of reduced robustness to noise, since the constellation points become closer together, making the system more sensitive to additive noise and channel distortions.



- In contrast, low-order modulations such as BPSK and QPSK provide better resilience to noise due to the larger Euclidean distance between constellation points, but they exhibit lower spectral efficiency.
- Converting a baseband signal to passband is achieved by modulating its in-phase and quadrature components with cosine and sine carriers, respectively. This translation shifts the spectrum around the carrier frequency, enabling transmission over RF channels while preserving the information carried by the baseband complex envelope.
- In practice, working in baseband is advantageous for design and analysis since it avoids redundant frequency components and simplifies filtering. However, passband signals are required when transmitting through real physical media that operate at specific frequency bands, such as wireless or optical communication systems. In addition, to obtain the modulated information, it is necessary to bring the signal back to baseband.

References

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